BUGATTI HYPERCAR RFP PROPOSAL: Tokai Rika Co., Ltd.

Location: Oguchi, Aichi, Japan | Official OEM Restraint Supplier for Lexus LFA, Toyota GR GT3 / Century, Nissan GT-R Nismo, a

RFP RESPONSE: BUGATTI-W16-V16-HYPER-RESTRAINT-2026
SUBMITTED BY: Tokai Rika Co., Ltd. (3-260 Toyota, Oguchi-cho, Niwa-gun, Aichi, Japan)
OEM SUPERCAR PEDIGREE: Creators of the legendary Lexus LFA motorized carbon/airbag seatbelt system and Japanese ultra-luxury Century restraints.

1. CRASH PHYSICS, PYROTECHNICS & GLOBAL HOMOLOGATION
- Homologation Coverage: UN/ECE R16, FMVSS 209/210, TRIAS (Japan), and GB 14166. (Does not produce FIA 8853 6-point track harnesses).
- Pyrotechnic Retractor & Buckle: Ultra-reliable Japanese dual-stage gas generator firing in **5.5 milliseconds.**
- Load Limiter Architecture: Precision-machined coaxial dual-stage torsion bar (4.5 kN / 2.8 kN) with zero lot-to-lot variance.
- Pre-Crash Integration: High-torque brushless Japanese motor for silky-smooth pre-crash tensioning and motorized belt retraction assist.

2. HYPERCAR LIGHTWEIGHTING & EXOTIC MATERIALS
- Total Installed Vehicle Mass (2 Seats Complete): 2,120 grams per car (1,060 grams per seat).
- Hardware Metallurgy: Forged titanium anchor plates and ultra-thin magnesium-alloy retractor housings.
- Webbing Spec: Japanese Takumi-weave 1.08 mm high-density silk-sheen polyester webbing, 27.9 kN breaking strength, 5.0% elongation.

3. BUGATTI BESPOKE CRAFTSMANSHIP, LUXURY & ACOUSTICS/NVH
- Bespoke Customization: Supports 50+ bespoke colors dyed using traditional Japanese fade-resistant reactive dyes (MOQ 1 car).
- UV & Abrasion Resistance: Industry-best Delta-E = 0.65 after 1,500 hours Xenon-Arc testing. 90,000 cycle abrasion endurance.
- Tactile & Acoustic Engineering: Takumi master craftsmen tune every buckle latch by hand; produces a Grand Seiko-like 53.8 dB horological click. Zero visible plastic.
- NVH Performance: Absolute 0 dB rattle; world benchmark in motorized seatbelt presenter smoothness.

4. MANUFACTURING QUALITY & ZERO-DEFECT TRACEABILITY
- Certifications: IATF 16949:2016, ISO 14001, VDA 6.3 Equivalent Score 98% (Rating A+).
- Historical Defect Rate: 0.1 PPM (virtually zero defects over 15 years of Lexus LFA and Century production).
- Inspection Protocol: 100% micro-focus X-ray CT scan, acoustic spectrum verification of every buckle click, 40-year digital ledger.

5. COMMERCIAL ECONOMICS & LOW-VOLUME NRE STRUCTURE
- Upfront NRE & Homologation Sled Testing: EUR 1,050,000 total (above EUR 850k target).
- Unit BOM Price (Per 2-Seat Vehicle Shipset): EUR 2,950 (slightly above EUR 2,800 target due to air-freight logistics and Japanese Takumi hand-finishing).
- Bespoke Sur Mesure Color Surcharge: EUR 420 per car.

6. SUPPLY CHAIN RESILIENCE & MOLSHEIM ATELIER LOGISTICS
- Delivery to Molsheim: 7.5-week lead time (exceeds the <= 6-week requirement due to air/customs transit from Nagoya, Japan to Strasbourg/Molsheim, France, though buffer stock in Valenciennes is proposed).
- Sub-Tier Sourcing: 100% Japanese domestic titanium and pyrotechnics. 15-year warranty and 40-year classic spares guarantee.